In [ ]:
```
EXPERIMENT 2 – NUMPY OPERATIONS
PIN NO: 241UDS0109

AIM:
To study and practice NumPy operations including array creation, attributes, indexing, slicing, broadcasting,
arithmetic, statistical operations, concatenation, reshaping, sorting, and splitting.
```

In [ ]:
```
## Step 1 – Install and Import NumPy
If NumPy is not installed, run `%pip install numpy pandas` in a separate cell first.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2 – Create Arrays for Three Energy Sources
Each array contains energy generated in MWh for Week 1 through Week 4.
```

In [2]:
```
solar_energy_241UDS0109 = np.array([840, 920, 875, 990])
wind_energy_241UDS0109 = np.array([610, 700, 660, 735])
hydro_energy_241UDS0109 = np.array([430, 455, 440, 480])

solar_energy_241UDS0109, wind_energy_241UDS0109, hydro_energy_241UDS0109
```

Out[2]:
```
(array([840, 920, 875, 990]),
 array([610, 700, 660, 735]),
 array([430, 455, 440, 480]))
```

In [ ]:
```
## Step 3 – Check Array Properties
```

In [3]:
```
solar_energy_241UDS0109.ndim
```

Out[3]:
```
1
```

In [4]:
```
wind_energy_241UDS0109.shape
```

Out[4]:
```
(4,)
```

In [5]:
```
hydro_energy_241UDS0109.size
```

Out[5]:
```
4
```

In [6]:
```
solar_energy_241UDS0109.dtype
```

Out[6]:
```
dtype('int64')
```

In [7]:
```
solar_energy_241UDS0109.itemsize, wind_energy_241UDS0109.itemsize, hydro_energy_241UDS0109.itemsize
```

Out[7]:
```
(8, 8, 8)
```

In [ ]:
```
## Step 4 – Retrieve Data Using Indexing and Slicing
```

In [8]:
```
hydro_energy_241UDS0109[0]
```

Out[8]:
```
np.int64(430)
```

In [9]:
```
np.unique(wind_energy_241UDS0109)
```

Out[9]:
```
array([610, 660, 700, 735])
```

In [10]:
```
solar_energy_241UDS0109[0:2]
```

Out[10]:
```
array([840, 920])
```

In [11]:
```
wind_energy_241UDS0109[2:]
```

Out[11]:
```
array([660, 735])
```

In [12]:
```
hydro_energy_241UDS0109[-3:]
```

Out[12]:
```
array([455, 440, 480])
```

In [ ]:
```
## Step 5 — Assign New Values Using Broadcasting
Copy the solar array, append two new values, then replace those appended values using slice assignment.
```

In [13]:
```
solar_energy_241UDS0109
```

Out[13]:
```
array([840, 920, 875, 990])
```

In [14]:
```
solar_energy_new_241UDS0109 = solar_energy_241UDS0109.copy()
solar_energy_new_241UDS0109
```

Out[14]:
```
array([840, 920, 875, 990])
```

In [15]:
```
solar_energy_new_241UDS0109 = np.append(solar_energy_new_241UDS0109, [1050, 970])
solar_energy_new_241UDS0109
```

Out[15]:
```
array([ 840,  920,  875,  990, 1050,  970])
```

In [16]:
```
solar_energy_new_241UDS0109[4:6] = [1080, 1010]
solar_energy_new_241UDS0109
```

Out[16]:
```
array([ 840,  920,  875,  990, 1080, 1010])
```

In [ ]:
```
## Step 6 — Arithmetic Operations
```

In [17]:
```
wind_energy_241UDS0109
```

Out[17]:
```
array([610, 700, 660, 735])
```

In [18]:
```
wind_energy_241UDS0109 + 25
```

Out[18]:
```
array([635, 725, 685, 760])
```

In [19]:
```
wind_energy_241UDS0109[3] - 15
```

Out[19]:
```
np.int64(720)
```

In [20]:
```
wind_energy_241UDS0109[0] * 2
```

Out[20]:
```
np.int64(1220)
```

In [21]:
```
wind_energy_241UDS0109[1:3] / 2
```

Out[21]:
```
array([350., 330.])
```

In [ ]:
```
## Step 7 — Statistical Operations
```

In [22]:
```
solar_energy_241UDS0109, wind_energy_241UDS0109, hydro_energy_241UDS0109
```

Out[22]:
```
(array([840, 920, 875, 990]),
 array([610, 700, 660, 735]),
 array([430, 455, 440, 480]))
```

In [23]:
```
np.sum(solar_energy_241UDS0109)
```

Out[23]:
```
np.int64(3625)
```

In [24]:
```
np.max(solar_energy_241UDS0109), np.min(solar_energy_241UDS0109)
```

Out[24]:
```
(np.int64(990), np.int64(840))
```

In [25]:
```
print(np.mean(solar_energy_241UDS0109), np.mean(wind_energy_241UDS0109), np.mean(hydro_energy_241UDS0109))
```

```
906.25 676.25 451.25
```

In [ ]:
```
## Step 8 — Combine All Energy-Source Data
```

In [26]:
```
solar_energy_241UDS0109.ndim, wind_energy_241UDS0109.ndim, hydro_energy_241UDS0109.ndim
```

Out[26]:
```
(1, 1, 1)
```

In [29]:
```
np.sum(all_energy_241UDS0109)
```

Out[29]:
```
np.int64(8135)
```

In [ ]:
```
## Step 9 — Reshape into a 3 × 4 Matrix
```

In [30]:
```
all_energy_241UDS0109.ndim, all_energy_241UDS0109.shape
```

Out[30]:
```
(1, (12,))
```

In [31]:
```
all_energy_241UDS0109 = all_energy_241UDS0109.reshape(3, 4)
all_energy_241UDS0109
```

Out[31]:
```
array([[840, 920, 875, 990],
       [610, 700, 660, 735],
       [430, 455, 440, 480]])
```

In [32]:
```
all_energy_241UDS0109.ndim, all_energy_241UDS0109.shape
```

Out[32]:
```
(2, (3, 4))
```

In [ ]:
```
## Step 10 — Sort and Split the Array
```

In [33]:
```
all_energy_241UDS0109
```

Out[33]:
```
array([[840, 920, 875, 990],
       [610, 700, 660, 735],
       [430, 455, 440, 480]])
```

In [34]:
```
all_energy_241UDS0109[0]
```

Out[34]:
```
array([840, 920, 875, 990])
```

In [35]:
```
np.sort(all_energy_241UDS0109, axis=1)
```

Out[35]:
```
array([[840, 875, 920, 990],
       [610, 660, 700, 735],
       [430, 440, 455, 480]])
```

In [36]:
```
np.sort(all_energy_241UDS0109, axis=0)
```

Out[36]:
```
array([[430, 455, 440, 480],
       [610, 700, 660, 735],
       [840, 920, 875, 990]])
```

In [ ]:
```
## Step 11 — Split the Array
```

In [37]:
```
split_solar_241UDS0109, split_wind_241UDS0109, split_hydro_241UDS0109 = np.split(
    all_energy_241UDS0109,
    [1, 2],
    axis=0
)
```

In [38]:
```
split_solar_241UDS0109
```

Out[38]:
```
array([[840, 920, 875, 990]])
```

In [39]:
```
split_wind_241UDS0109
```

Out[39]:
```
array([[610, 700, 660, 735]])
```

In [40]:
```
split_hydro_241UDS0109
```

Out[40]:
```
array([[430, 455, 440, 480]])
```

In [ ]:
```
## Step 12 — Retrieve Week 2 Energy Generation of All Sources
```

In [41]:
```
all_energy_241UDS0109
```

Out[41]:
```
array([[840, 920, 875, 990],
       [610, 700, 660, 735],
       [430, 455, 440, 480]])
```

In [42]:
```
week_2_energy_241UDS0109 = all_energy_241UDS0109[:3, 1:2]
print(week_2_energy_241UDS0109)
```

```
[[920]
 [700]
 [455]]
```

In [ ]:
```
## RESULT
NumPy arrays were created for three renewable-energy sources, and array properties, indexing, slicing, value
assignment, arithmetic, statistical operations, concatenation, reshaping, sorting, splitting, and Week 2 selection
were performed.
```